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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

AD2105-1- BIG DATA ANALYTICS
(For AD)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which of the following is an example of structured data?
A) Emails
B) Social media posts
C) Audio files
D) SQL database records
2. What is a key characteristic of unstructured data?
A) Fixed schema
B) Easily searchable
C) Lacks a predefined format
D) Stored in relational databases
3. Which of the following is a characteristic of Big Data?
A) Volume, Variety, and Velocity
B) Structure, Simplicity, and Speed
C) Complexity, Accuracy, and Cost
D) Storage, Security, and Scalability
4. What is the main function of the MapReduce framework?
A) Storing data
B) Processing large datasets
C) Visualizing data
D) Securing data
5. What does DAG stand for in Apache Spark?
A) Directed Acyclic Graph
B) Data Analysis Group
C) Distributed Array Grid
D) Dynamic Allocation Graph
6. What is lazy evaluation in Apache Spark?
A) Executing operations immediately
B) Delaying execution until an action is triggered
C) Reducing memory usage
D) Optimizing data storage
7. Which compression format is commonly used in Big Data environments?
A) PNG
B) GZIP
C) JPEG
D) MP4
8. What does functional programming emphasize?
A) Iterative loops
B) Object creation
C) Immutable data and pure functions
D) Dynamic typing
9. What is the primary function of HDFS in Hadoop?
A) Data processing
B) Data storage
C) Data visualization
D) Data encryption
10. Which component of Hadoop is responsible for managing the file system namespace?
A) DataNode
B) NameNode
C) JobTracker
D) TaskTracker
11. Which of the following is a feature of Hadoop streaming?
A) Allows running non-Java programs
B) Compresses data automatically
C) Provides a SQL interface
D) Manages metadata

12. What is Avro used for in Hadoop?
 A) Data compression B) File-based data structures
 C) Task scheduling D) Data encryption
13. Which programming language is primarily used to write HBase clients?
 A) Python B) Java
 C) Scala D) C++
14. Which command-line tool is used to interact with Pig?
 A) Grunt B) Bash
 C) HiveCLI D) SparkShell
15. Which file format is commonly used in Hive?
 A) JSON B) Parquet
 C) CSV D) HTML
16. What is the purpose of the Partitioner in MapReduce?
 A) Distribute data across nodes B) Sort intermediate keys
 C) Split input data D) Manage data integrity
17. What is a DataFrame in Spark?
 A) A distributed collection of data organized into named columns B) A single node database
 C) A visualization tool D) A data compression technique
18. Which Spark action returns the number of elements in an RDD?
 A) filter B) map
 C) count D) reduce
19. Which query language does MongoDB use?
 A) SQL B) MongoDB Query Language (MQL)
 C) Pig Latin D) SparkQL
20. What is the primary role of Spark Core?
 A) Handle SQL queries B) Provide streaming capabilities
 C) Handle basic distributed data processing D) Offer machine learning libraries

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the classification of digital data. How is structured data different from unstructured data? Provide examples.	8	L*2	1	1
	b) Explain the Hadoop ecosystem and its core components. How does it support Big Data processing?	8	L2	2	1
2.	a) Explain functional programming in Scala. How do higher-order functions and tail recursion enhance the efficiency of Scala programs?	8	L2	1	1
	b) What is pattern matching in Scala? Explain its syntax and usage with suitable examples.	8	L2	2	1
3.	a) What are the key terminologies used in Big Data environments? Explain each term with its significance in Big Data processing.	8	L2	3	1
	b) Explain the Directed Acyclic Graph (DAG) execution model in Apache Spark. How does it improve the performance of Big Data processing?	8	L2	3	1
Unit – II					
4.	a) Explain the data flow in HDFS. How does Hadoop handle read and write operations in a distributed environment?	8	L2	4	1
	b) Explain how data analysis is performed with Hadoop. What are the key tools and techniques used for analyzing large datasets?	8	L2	4	1

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| 5. | a) | Explain the HBase data model and its key components. How does HBase differ from traditional relational databases? | 8 | L2 | 5 | 1 |
| | b) | Explain the integration of Hadoop with Cassandra. How does this integration help in handling large-scale data storage and retrieval? | 8 | L2 | 5 | 1 |
| 6. | a) | Explain the Hive query language (HiveQL). How does it facilitate data definition, manipulation, and querying in Hadoop? | 8 | L2 | 5 | 1 |
| | b) | What are the key features of Pig in Hadoop? Explain the Pig data model and the role of Pig Latin in Big Data processing. | 8 | L2 | 5 | 1 |

Unit – III

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| 7. | a) | Explain the differences between DataFrames and Datasets in Apache Spark. How do they improve performance? | 8 | L2 | 4 | 1 |
| | b) | Explain Apache Spark SQL and its role in processing structured data. How does it integrate with DataFrames and Datasets? | 8 | L2 | 4 | 1 |
| 8. | a) | Explain the core modules of Apache Spark. How do key-value pairs enhance the functionality of RDDs in distributed computing? | 8 | L2 | 4 | 1 |
| | b) | Explain MongoDB query language with examples. How does MongoDB handle different data types and support complex queries? | 8 | L2 | 4 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
